

- **Maximum likelihood (ML)** is usually the default method in most programs because it yields the most precise (smallest variance) estimates when the data are normal.
- **Generalized least squares (GLS)** has the same optimal properties as ML under normality.
- The **asymptotically distribution free (ADF)** method has no distributional assumptions and hence is most general, but it is impractical with many variables and inaccurate without very large sample sizes.

Satorra and Bentler (1988, 1994, and 2001) and Satorra (2000) also developed an adjustment for non-normality that can be applied to the ML, GLS, or EDT chi-square test statistics. Briefly, the Satorra–Bentler scaled χ^2 is a correction to the χ^2 test statistic (Satorra & Bentler, 2001). EQS also corrects the standard errors for parameter estimates to adjust for the extent of non-normality (Bentler & Dijkstra, 1985).

Sample size and plausibility of the normality and independent assumptions must be considered while choosing the suitable estimation technique. ML, the Scaled ML, or GLS estimators may be good choices with medium (over 120) to large samples and evidence of the plausibility of the normality assumptions. ML estimation is currently the most frequently used estimation method in SEM. In medium (over 120) to large samples the scaled ML test statistic is a good choice with non-normality or suspected dependence among factors and errors. In small samples (60 to 120) the Yuan–Bentler test statistic appears best. The test statistic based on the ADF estimator (without adjustment) is inappropriate choice under all conditions unless the sample size is very large (> 2,500).

17.10.3 Model Evaluation

A central issue addressed by SEM is how to assess the fit between observed data and the hypothesized model ideally operationalized as an evaluation of the degree of discrepancy between the true population covariance matrix and that implied by the model's structural and non structural parameters. As the population parameter values are rarely identified, the difference between an observed, sample-based covariance matrix and that implicit by parameter estimates must serve to approximate the population discrepancy. The observed data will fit the model perfectly for a just identified model. The system of equations expressing each model parameter as a function of the observed (co)variances is uniquely solvable; thus, the sample estimate of the model-implied covariance matrix will, by default, equal the sample estimate of the population covariance matrix. However, if a model is over identified, it is unlikely that these two matrices are equal as the system of equations (expressing model parameters as functions of observed variances and covariances) is solvable in more than a single way.

Abiding by a universal desire for parsimony, over identified models tend to be of more substantive interest than just identified ones because they characterize simpler potential explanations of the observed associations. While data-model fit for such models was initially conceived as a formal statistical test of the discrepancy between the true and model-implied covariance matrices, such a test now is often viewed as overly strict given its power to detect even trivial deviations of a proposed model from reality. Hence, many alternative

assessment strategies have emerged and continue to be developed. Data-model fit indices for such assessment can be categorized roughly into three broad classes:

- **Absolute indices** assess the overall discrepancy between observed and implied covariance matrices; fit improves as more parameters are added to the model and degrees of freedom decrease: for example, the standardized root mean square residual (SRMR), the chi-square test (recommended to be reported mostly for its historical significance), and the goodness-of-fit index (GFI).
- **Parsimonious indices** assess the overall discrepancy between observed and implied covariance matrices while taking into consideration a model's complexity; fit improves as more parameters are added to the model, as long as those parameters are making a useful contribution: for example, the root mean square error of approximation (RMSEA) with its associated confidence interval, the Akaike information criterion (AIC) for fit comparisons across non-nested models, and the adjusted goodness-of-fit index (AGFI).
- **Incremental indices** assess absolute or parsimonious fit relative to a baseline model, usually the null model (a model that specifies no relations among measured variables): for example, the comparative fit index (CFI), the normed fit index (NFI), and the non-normed fit index (NNFI).

SEM software programs routinely report a handful of goodness-of-fit indices. Some of these indices work better than others under certain conditions. It is generally recommended that multiple indices be considered simultaneously when overall model fit is evaluated. For instance, Hu and Bentler (1999) proposed a 2-index strategy that is, reporting SRMR along with one of the fit indices (e.g., RNI, CFI, or RMSEA). The authors also suggested the following criteria for an indication of good model-data fit using those indices: $RNI \text{ (or CFI)} \geq .95$, $SRMR \leq .08$, and $RMSEA \leq .06$. Despite the sample size sensitivity problem with the chi-square test, reporting the model chi-square value with its degrees of freedom in addition to the other fit indices is recommended. If, after considering several indices, data model fit is deemed acceptable (and judged best compared to competing models, if applicable), the model is reserved as rational, and individual parameters may be interpreted. If, however, evidence suggests unacceptable data-model fit, the next and often final stage in the SEM process is considered: modifying the model to improve fit in hopes of also improving the model's correspondence to reality.

17.10.4 Model Modification

Any hypothesized model is only an approximation to reality; the question remains is one of degree of that misspecification. With regard to external specification errors—when irrelevant variables are included in the model or substantively important ones remain left out—remediation can only occur by respecifying the model based on more relevant theory. On the other hand, internal specification errors—when unimportant paths among variables were included or when important paths were omitted—can potentially be diagnosed and remedied using Wald statistics (predicted increase in chi-square if a previously estimated parameter were fixed to some known value, e.g., zero) and Lagrange multiplier statistics (also referred to as modification indices; estimated decrease in chi-square if a previously fixed parameter were to be estimated). All are asymptotically equivalent under the null hypothesis but

approach model modification differently. As these tests' recommendations are directly motivated by the data and not by theoretical considerations, any resulting respecifications must be viewed as exploratory in nature and might not lead to a model that resembles reality any more closely than the one(s) initially conceptualized.

Chi-square difference test

If models are nested, that is, models are subsets of each other, the χ^2 value for the larger model is subtracted from the χ^2 value for the smaller, nested model and the difference, also χ^2 , is evaluated with degrees of freedom equal to the difference between the degrees of freedom in the two models. When the data are normally distributed, the chi-squares can simply be subtracted. However, when the data are non normal and the Satorra–Bentler scaled chi-square is employed, an adjustment is required (Satorra, 2000; Satorra & Bentler, 2001) so that the S–B χ^2 difference test is distributed as a chi-square.

Chi-square difference tests are popular methods for comparison of nested models especially when, as in this example, there are two nested a priori models of interest; however, a potential problem arises when sample sizes are small. Because of the relationship between sample size and χ^2 , it is hard to detect a difference between models with small sample sizes.

Lagrange Multiplier (LM) test

The LM test also compares nested models but requires estimation of only one model. The LM test asks would the model be improved if one or more of the parameters in the model that are currently fixed were estimated. Or, equivalently, what parameters should be added to the model to improve the fit of the model?

There are many approaches to using the LM tests in model modifications. It is possible and often desirable to look only at certain parts of the model for possible change although it is also possible to examine the entire model for potential additions.

The LM test can be examined either univariately or multivariately. There is a danger in examining only the results of univariate LM tests because overlapping variance between parameter estimates may make several parameters appear as if their addition would significantly improve the model. All significant parameters are candidates for inclusion by the results of univariate LM tests, but the multivariate LM test identifies the single parameter that would lead to the largest drop in the model χ^2 and calculates the expected change in χ^2 if this parameter was added. After this variance is removed, the next parameter that accounts for the largest drop in model χ^2 is assessed, similarly. After a few candidates for parameter additions are identified, it is best to add these parameters to the model and repeat the process with a new LM test, if necessary.

Wald test

While the LM test asks which parameters, if any, should be added to a model, the Wald test asks which, if any, could be deleted. Are there any parameters that are currently being estimated that could instead be fixed to zero? Or, equivalently, which parameters are not necessary in the model? The Wald test is analogous to backward deletion of variables in stepwise regression, in which one seeks a non significant change in R^2 when variables are left out. However, if the goal had been the development of a parsimonious model, the Wald test could have been examined to evaluate deletion of unnecessary parameters.

Finally, no matter how complex the model under study, it is always conceivable that other variables or causal assumptions that the researcher had failed to consider can account for the observed relationships. To enhance the theoretical value of their investigations, researchers are well advised to consider also models that are almost equivalent to the one reported. These are models that fit the analyzed data set almost as well as the reported model and that may offer interesting alternative theoretical interpretations.

Comparison of Nested Models

In addition to evaluating the overall model fit and specific parameter estimates, it is also possible to statistically compare nested models to one another. Nested models are models that are subsets of one another. When theories can be specified as nested hypotheses, each model might represent a different theory. These nested models are statistically compared, thus providing a strong test for competing models. Some of the common variance among the items may be due to wording and the general domain area (e.g., health problems, relationships problems), not solely due to the underlying constructs. We can compare our model to a model that also includes paths that account for the variance explained by the general domain or wording of the item. The model with the added paths to account for this variability would be considered the full model. To test this hypothesis, the chi-square from the model with paths added to account for domain and wording would be subtracted from the chi-square for the nested model in that does not account for common domains and wording among the items. The corresponding degrees of freedom for these two models would also be subtracted. Given nested models and normally distributed data, the difference between two chi-squares is a chi-square with degrees of freedom equal to the difference in degrees of freedom between the two models. The significance of the chi-square difference test can then be assessed in the usual manner. If the difference is significant, the fuller model that includes the extra paths is needed to explain the data. If the difference is not significant, the nested model, which is more parsimonious than the fuller model, would be accepted as the preferred model.

17.10.5 Reporting the Results

The final step is to precisely and completely describe the analysis in written reports. How authors structure the results section obviously is dictated by the particular model(s) and research questions under study. It is the researcher's responsibility to provide access to data in order to facilitate verification of the obtained results: If moment-level data were analyzed, a covariance matrix (or correlation matrix with standard deviations) should be offered in a table or appendix; if raw data were used, information on how to obtain access should be provided.

Results from both the measurement and structural phases should be presented. Judging the overall quality of a hypothesized model usually is presented early in the results section or overidentified models. Given that available data-model fit indices can lead to inconsistent conclusions, researchers should consider fit results from different classes so readers can arrive at a more complete picture regarding an model's acceptability. Also, a comparison of fit across multiple, a priori specified alternative models can help in weighing the relative merits of preferring one model over others. As illustrated, when competing models are nested, a formal chi-square difference test is available to judge if a more restrictive-but also more parsimonious-model can elucidate the observed data equally well, without a considerable loss in data-model fit (alternative models that

are not nested have traditionally been compared only descriptively—relative evaluations of values are recommended, with smaller values indicating better fit—but recent methodological developments suggest statistical approaches as well).

If post hoc model modifications are performed following unacceptable data-model fit from either the measurement or structural phase of the analysis, authors should give a thorough account of the nature and reasons (both statistical and theoretical) for the respecification(s), including summary results from Lagrange multiplier tests and revised final fit results. If data-model fit has been assessed and believed to be satisfactory, with or without respecification, more detailed results are offered, usually in the form of individual unstandardized and standardized parameter estimates for each structural equation of interest, together with associated standard errors and/or test statistics and coefficients of determination. When latent variables are part of a model, estimates of their construct reliability should be presented, with values ideally falling above .70 or .80.

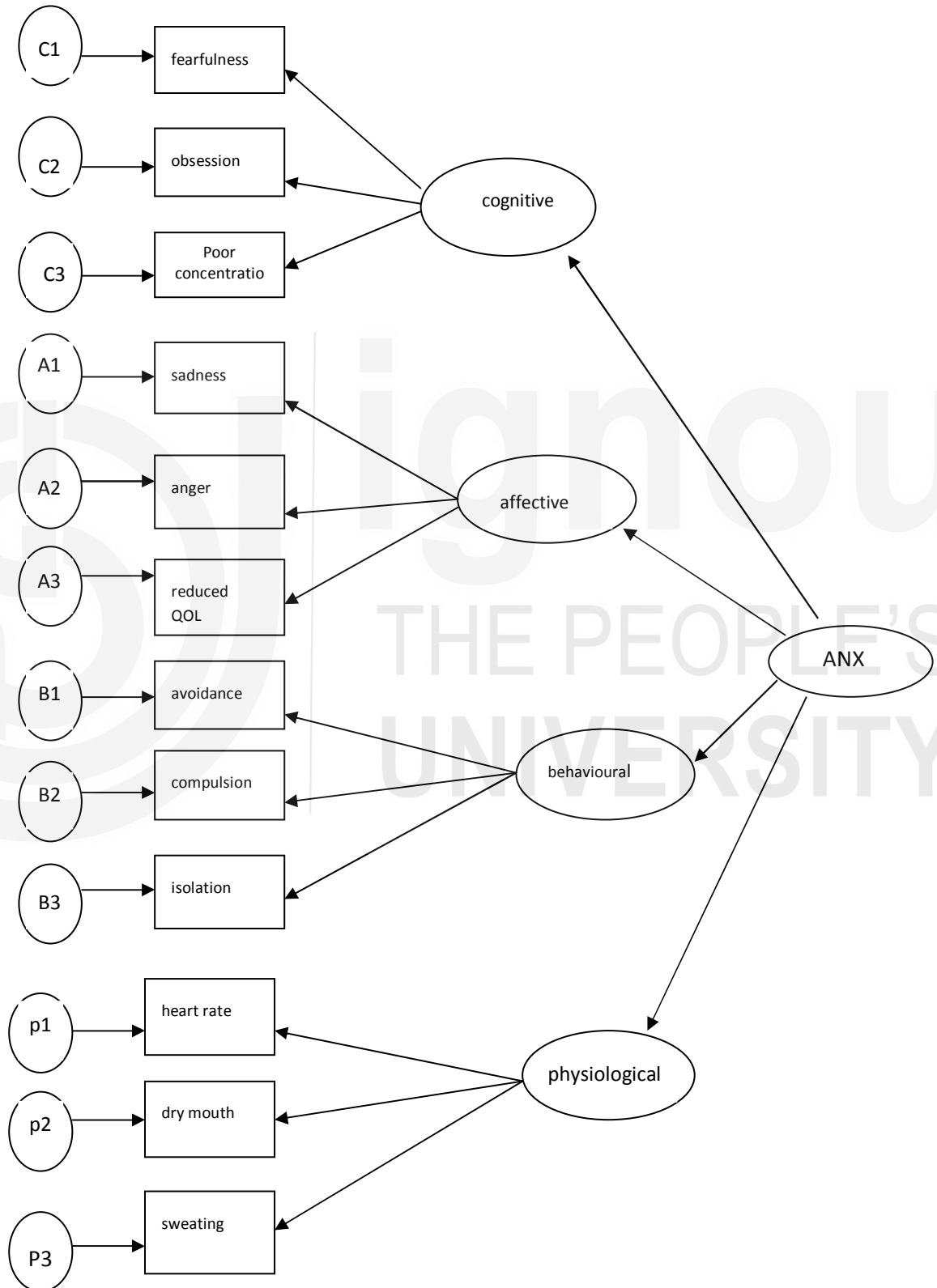
Researcher should provide a logic of what implications the results from the SEM analysis have on the theory or theories that gave rise to the initial model(s). Claims that a well-fitting model was “confirmed” or that a particular theory was proven to be “true,” especially after post hoc respecifications, should be avoided. Such statements are greatly deceptive given that alternative, structurally different, yet mathematically equivalent models always exist that would produce identical data-model fit results and thus would explain the data equally well. At most, a model with acceptable fit may be interpreted as *one* reasonable explanation for the associations observed in the data. From this perspective, a SEM analysis should be evaluated from a disconfirmatory, rather than a confirmatory perspective. Based on unacceptable data model fit results, theories can be shown to be false but not proven to be true by acceptable data-model fit.

If evidence of data-model misfit was presented and a model was modified based on statistical results from Lagrange multiplier tests, readers must be made aware of potential model overfitting and the capitalization on chance. Statistically rather than theoretically based respecifications are purely exploratory and might say little about the true model underlying the data. While some model modifications seem appropriate and theoretically justifiable (usually, minor respecifications of the measurement portion are more easily defensible than those in the structural portion of a model), they only address internal specification errors and should be cross-validated with data from new and independent samples. Finally, the interpretation of individual parameter estimates can involve explicit causal language, as long as this is done from within the context of the particular causal theory proposed and the possibility/probability of alternative explanations is raised unequivocally. Though some might disagree, we think that explicit causal statements are more honest than implicit ones and are more useful in articulating a study’s practical implications; after all, is not causality the ultimate aim of science.

In the end, SEM is a powerful disconfirmatory tool at the researcher’s disposal for testing and interpreting theoretically derived causal hypotheses from within an a priori specified causal system of observed and/or latent variables. However, we urge authors to resist the apparently still popular belief that the main goal of SEM is to achieve satisfactory data-model fit results; rather, it is to get one step closer to the “truth.” If it is true that a proposed model does not reflect reality, then reaching a conclusion of misfit between data and model should be a desirable goal, not one to be avoided by careless respecifications until satisfactory levels of fit are achieved.

17.11 AN EXAMPLE OF SEM

For example, we take some fictitious data in which anxiety was measured. We performed an confirmatory factor analysis in which we defined anxiety as consisting of four components, namely, affective, behavioral, cognitive and physiological symptoms. Thus, we specified a model which looked like as the figure below:



The next step is to estimate the parameters of the model. The software performs this step and provides final results of parameters that produce best fit to data possible.

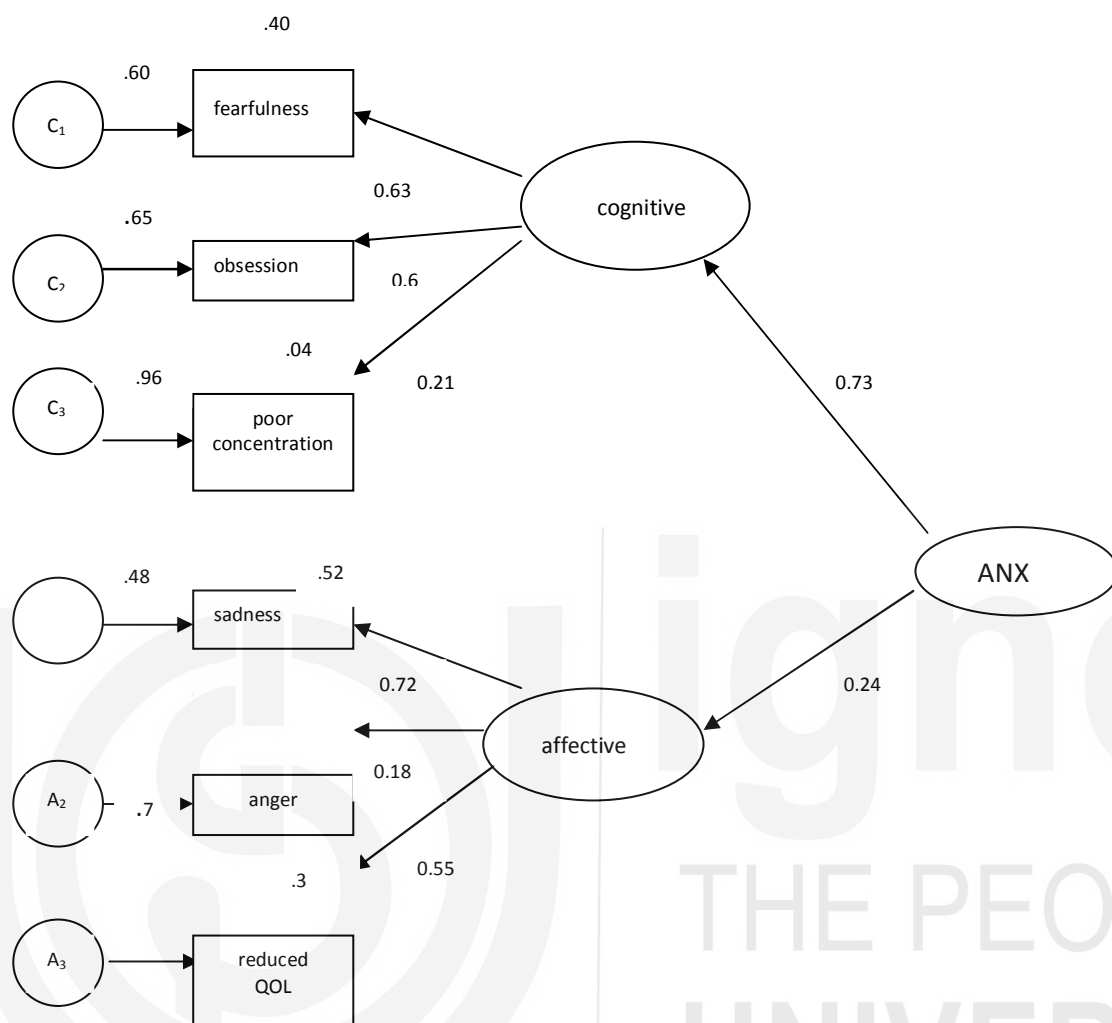


Figure 2 shows only 2 of the 4 components of anxiety. Above each arrow from a latent variable to a variable is a number, called the path coefficient. This can range from -1.0 to 1.0 , with higher numbers (positive or negative) showing a stronger association. As can be seen, the variable “obsessions” doesn’t fit very well with the cognitive trait; and the variable “anger” doesn’t seem to go with the affective trait. The numbers over the variable names are the SMRs, which are the squared values of the path coefficients; they are interpreted in the same way as R^2 multiple regression—in terms of how much of the variance in one variable is explained by, or are in common with, the other variable. Finally, the numbers over the arrows between the error terms and the variables are the variances of the errors. You’ll note that the sum of the SMR plus the error variance for each variable is 1.0 ; that is, all the variance of a variable is divided between that shared with the latent variable and error. The cognitive domain is correlated more highly with the latent trait of anxiety than is the affective realm, as reflected in their respective path coefficients.

The next step is model specification stage. Although no mathematics is involved, it is probably the most difficult—and most important—part. It is the most difficult because it requires the most thought and understanding of the

theoretical model of the purported influences. No computer program can help us at this stage, only our knowledge of the field. It is the most important step because everything depends on how well we specify the model. The computer programs may help us in determining whether some variables aren't important, we can play with different paths to see whether they improve the model or not. However, the primary cause of poorly fitting is the omission of crucial variables.

The next step is relatively easy. The main focus is on the paths—from the latent variable to the measured variables and from that latent variable to the next one in the path. Once we have cleaned up the model by pruning noncontributory paths, we can examine the fit of the overall model. The most common index of how well the data match the model is RMSEA. It sees how much the data deviate from the model. Values over 0.10 are considered to be a bad fit, those less than 0.08 reflect a reasonable fit, and values less than 0.05 indicate a good fit.

Given is the table showing RMSEA of the model:

RMSEA

Model	RMSEA	LO 90	HI 90	<u>PCLOSE</u>
Independence model	.100	.087	.115	.000

PCLOSE = .000 for the **Independence model**. Under the hypothesis of “close fit” (i.e., that RMSEA is no greater than .05 in the population), the probability of getting a sample RMSEA as large as .100 is .000.

17.12 SOFTWARE PROGRAMS FOR SEM

Most SEM analyses are conducted using one of the specialized SEM software programs. However, there are many options, and the choice is not always easy. Below is a list of the commonly used programs for SEM. Special features of each program are briefly discussed. It is important to note that this list of programs and their associated features is by no means comprehensive. This is a rapidly changing area and new features are regularly added to the programs.

17.12.1 LISREL

LISREL (linear structural relationships) is one of the earliest SEM programs and perhaps the most frequently referenced program in SEM articles. Its version 9.1 has three components: PRELIS, SIMPLIS, and LISREL. PRELIS (pre-LISREL) is used in the data preparation stage when raw data are available. Its main functions include checking distributional assumptions, such as univariate and multivariate normality, imputing data for missing observations, and calculating summary statistics, such as Pearson covariances for continuous variables, polychoric or polyserial correlations for categorical variables, means, or asymptotic covariance matrix of variances and covariances (required for asymptotically distribution-free estimator or Satorra and Bentler's scaled chi-square and robust standard errors). PRELIS can be used as a stand-alone program or in conjunction with other programs.

Summary statistics or raw data can be read by SIMPLIS or LISREL for the estimation of SEM models. The LISREL syntax requires understanding of

matrix notation while the SIMPLIS syntax is equation-based and uses variable names defined by users. Both LISREL and SIMPLIS syntax can be built through interactive LISREL by entering information for the model construction wizards. Alternatively, syntax can be built by drawing the models on the Path Diagram screen.

LISREL 9.1 allows the analysis of multilevel models for hierarchical data in addition to the core models. A free student version of the program, which has the same features as the full version but limits the number of observed variables to 12, is available from the web site of Scientific Software International, Inc. (<http://www.ssicentral.com>). This web site also offers a list of illustrative examples of LISREL's basic and new features.

17.12.2 EQS

Version 6.1 of EQS (Equations) provides many general statistical functions including descriptive statistics, *t*-test, ANOVA, multiple regression, nonparametric statistical analysis, and EFA. Various data exploration plots, such as scatter plot, histogram, and matrix plot are readily available in EQS for users to gain intuitive insights into modelling problems. Similar to LISREL, EQS allows different ways of writing syntax for model specification.

The program can generate syntax through the available templates under the "Build_EQS" menu, which prompts the user to enter information regarding the model and data for analysis, or through the Diagrammer, which allows the user to draw the model. Unlike LISREL, however, data screening (information about missing pattern and distribution of observed variables) and model estimation are performed in one run in EQS when raw data are available. Model-based imputation that relies on a predictive distribution of the missing data is also available in EQS. Moreover, EQS generates a number of alternative model chi-square statistics for non normal or categorical data when raw data are available. The program can also estimate multilevel models for hierarchical data. Visit <http://www.mvsoft.com> for a comprehensive list of EQS's basic functions and notable features.

17.12.3 Mplus

Version 7.3 of the Mplus program includes a Base program and three add-on modules. The Mplus Base program can analyze almost all single-level models that can be estimated by other SEM programs. Unlike LISREL or EQS, Mplus version 7.33 is mostly syntax-driven and does not produce model diagrams. Users can interact with the Mplus Base program through a language generator wizard, which prompts users to enter data information and select the estimation and output options. Mplus then converts the information into its program-specific syntax. However, users have to supply the model specification in Mplus language themselves.

Mplus Base also offers a robust option for non-normal data and a special full-information maximum likelihood estimation method for missing data (see footnote 4). With the add-on modules, Mplus can analyze multilevel models and models with latent categorical variables, such as latent class and latent profile analysis. The modelling of latent categorical variables in Mplus is so far unrivalled by other programs. The official web site of Mplus (<http://www.statmodel.com>) offers a comprehensive list of resources including

basic features of the program, illustrative examples, online training courses, and a discussion forum for users.

17.12.4 Amos

Amos (Analysis of Moment Structure) version 22 is distributed with IBM SPSS. It has two components: Amos Graphics and Amos Basic. Similar to the LISREL Path Diagram and SIMPLIS syntax, respectively, Amos Graphics permits the specification of models by diagram drawing whereas Amos Basic allows the specification from equation statements. A notable feature of Amos is its capability for producing bootstrapped standard error estimates and confidence intervals for parameter estimates.

An alternative full-information maximum likelihood estimation method for missing data is also available in Amos. The program is available at <http://www.smallwaters.com> or <http://www.spss.com/amos/>.

17.12.5 Mx

Mx (Matrix) version 6 is a free program downloadable from <http://www.vcu.edu/mx/>. The Mx Graph version is for Microsoft Windows users. Users can provide model and data information through the Mx programming language. Alternatively, models can be drawn in the drawing editor of the Mx Graph version and submitted for analysis. Mx Graph can calculate confidence intervals and statistical power for parameter estimates. Like Amos and Mplus, a special form of full-information maximum likelihood estimation is available for missing data in Mx.

17.12.6 Others

In addition to SPSS, several other general statistical software packages offer built-in routines or procedures that are designed for SEM analyses. They include the CALIS (covariance analysis and linear structural equations) procedure of SAS (SAS Institute Inc., 2000; <http://www.sas.com/>), the RAMONA (Reticular Action Model Or Near Approximation) module of SYSTAT (Systat Software, Inc., 2002; <http://www.systat.com/>), and SEPATH (structural equation modeling and path analysis) of Statistica (StatSoft, Inc., 2003; <http://www.statsoft.com/products/advanced.html>).

17.13 ADVANTAGES AND DISADVANTAGES OF SEM

17.13.1 Advantages of SEM

- 1) SEM allows researcher to identify direct and indirect effects. Direct effects are principally the relationship between a dependent variable we are interested in explaining and an independent variable we think is causing or related to the dependent variable. Thus this establishes links directly from a cause to an effect. An indirect effect occurs when one variable goes through another variable on the way to some dependent or independent variable.
- 2) It is possible to have multiple indicators of a concept. With multiple regression analysis we only evaluate the effect of individual independent variables on the dependent variable. It creates fairly a simple path model